

Model-EF Core與ViewModel

- **9-1:在.NET Core Web專案加入EF Model**

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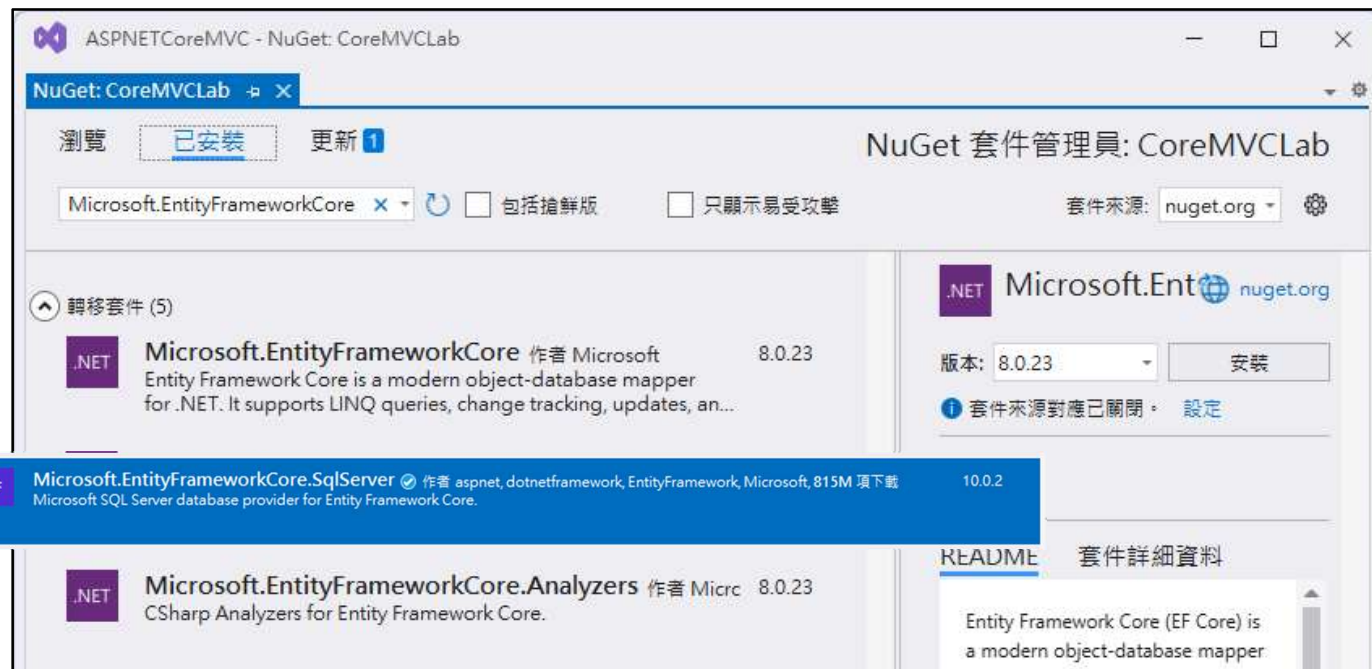
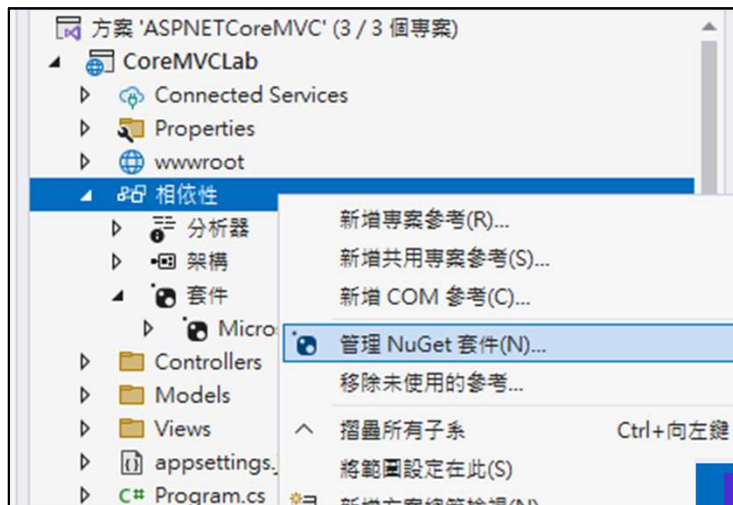
- **9-2:ViewModel**

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Module 09

安裝NuGet套件

- 開啟[管理NuGet套件]。
- 安裝「Microsoft.EntityFrameworkCore」與「Microsoft.EntityFrameworkCore.SqlServer」。



建立 DbSet Model

- 依據資料庫中的Table新建符合資料格式的Model。

```
public class Customer
{
    0 個參考
    public string CustomerID { get; set; }
    0 個參考
    public string CompanyName { get; set; }
    0 個參考
    public string? ContactName { get; set; }
    0 個參考
    public string? ContactTitle { get; set; }
    0 個參考
    public string? Address { get; set; }
    0 個參考
    public string? City { get; set; }
    0 個參考
    public string? Region { get; set; }
    0 個參考
    public string? PostalCode { get; set; }
    0 個參考
    public string? Country { get; set; }
    0 個參考
    public string? Phone { get; set; }
    0 個參考
    public string? Fax { get; set; }
}
```

9-3

張佑晨 編

建立 DbContext Model

```
public class NorthwindDbContext:DbContext
{
    0 個參考
    public NorthwindDbContext(DbContextOptions<NorthwindDbContext> options) : base(options) {}

    1 個參考
    public DbSet<Customer> Customers { get; set; }

    0 個參考
    protected override void OnModelCreating(ModelBuilder modelBuilder)
    {
        modelBuilder.Entity<Customer>(e =>
        {
            e.ToTable("Customers");    // 表名
            e.HasKey(x => x.CustomerID);    // 主鍵
            //e.Property(x => x.Phone).HasMaxLength(20).IsRequired(); // 屬性配置範例
        });
    }
}
```

Program.cs 設定連線字串與 DI

- 於Program.cs下加入DI設定。

```
builder.Services.AddDbContext<NorthwindDbContext>(options =>  
    options.UseSqlServer(builder.Configuration.GetConnectionString("Northwind")));
```

- 於appsettings.json設定連線字串。

```
"ConnectionStrings": {  
  "Default": "Server=.;Database=Northwind;Trusted_Connection=True;TrustServerCertificate=True"  
}
```

使用EF Core

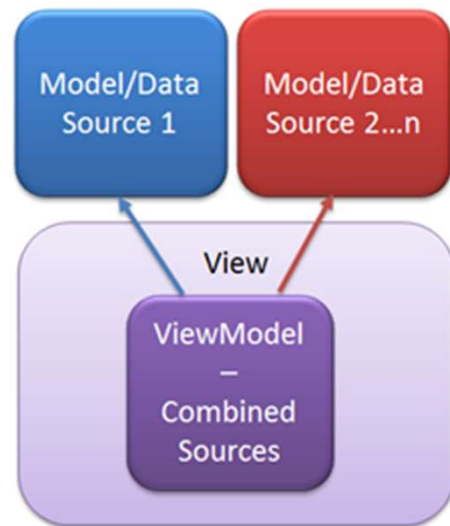
- 於Controller使用

```
public class CustomerController : Controller
{
    private readonly NorthwindDbContext _nwContext;
    0 個參考
    public CustomerController(NorthwindDbContext nwContext)
    {
        _nwContext = nwContext;
    }

    0 個參考
    public IActionResult GetAllCustomers()
    {
        var customers = _nwContext.Customers.ToList();
        return View(customers);
    }
}
```

ViewModel -1

- 與先前提到的Model相同都是Class類別，但ViewModel是專為View使用的Model。
- 可使多個不同Model資料，以一個Model呈現並共同繫結同個表單。
- 表單中若有欄位不需進入資料庫時可以使用（如僅當作狀態判斷）。
- ViewModel讓我們在View使用強型別資料繫結時更佳有彈性。



9-7

ViewModel -2

- 一般利用強型別的方式顯示資料(@model) :

```
@model IEnumerable<Store.Models.Members>
<h2>MemberList</h2>
@Html.ActionLink("新增會員資料", "CreateMemberForm", null, new { @class = "btn btn-secondary" })
@Html.ActionLink("以城市分類", "MemberByCity", null, new { @class = "btn btn-success" })
<table class="table table-striped table-hover">
    <thead>
        <tr class="row">
            <th class="col-sm-2">會員姓名</th>
            <th>Email</th>
            <th>城市</th>
            <th>會員描述</th>
            <th class="col-sm-1">詳細</th>
        </tr>
    </thead>
    <tbody>
        @foreach (var mb in Model)
        {
            <tr class="row">
                <td class="col-sm-2">@mb.MemberName</td>
                <td>@mb.Email</td>
                <td>@mb.CityID</td>
                <td>@mb.Description</td>
                <td class="col-sm-1">@Html.ActionLink("詳細", "MemberInfo", new { id = mb.MemberID })</td>
            </tr>
        }
    </tbody>
</table>
```


ViewModel -3

MemberList				
新增會員資料				
會員姓名	Email	城市	會員描述	編輯
John Wick	johnwick@jw.com	2	John is a man of focus, commitment, sheer will... something you know very little about. I once saw him kill three men in a bar... with a pencil. With a fucking pencil!"	詳細
張滿月	JangMan-wol@delluna.com	5	호텔 델루나 이지은	詳細
王雷	WangLei@ganlingnaa.com	7	早上買，中午吃，晚上走	詳細
諸葛村夫	farmer@shu.com	8	臣本布衣，躬耕於南陽，苟全性命於亂世，不求聞達於諸侯。先帝不以臣卑鄙，猥自枉屈，三顧臣於草廬之中，諮臣以當世之事，由是感激，遂許先帝以驅馳。後值傾覆，受任於敗軍之際，奉命於危難之間：爾來二十有一年矣。	詳細
陳浩南	AndyChan@teddyboy.com	6	銅鑼灣只有一個浩南，就是我陳浩南!	詳細

- 需要改善的內容。

ViewModel -4

- 將Members與City類別結合View要顯示的Model。

```
public partial class Members
{
    public int MemberID { get; set; }
    public string MemberName { get; set; }
    public string Email { get; set; }
    public string Password { get; set; }
    public int CityID { get; set; }
    public string Description { get; set; }
}
```



```
public partial class City
{
    public int CityID { get; set; }
    public string CityName { get; set; }
}
```



```
public class MembersViewModel
{
    public int MemberID { get; set; }
    public string MemberName { get; set; }
    public string Email { get; set; }
    public string Description { get; set; }

    public string CityName { get; set; }
}
```

ViewModel -5

```
@model IEnumerable<Store.Models.ViewModel.MembersViewModel>
<h2>MemberList</h2>
@Html.ActionLink("新增會員資料", "CreateMemberForm", null, new { @class = "btn btn-secondary" })
@Html.ActionLink("以城市分類", "MemberByCity", null, new { @class = "btn btn-success" })
<table class="table table-striped table-hover">
  <thead>
    <tr class="row">
      <th class="col-sm-2">會員姓名</th>
      <th>Email</th>
      <th>城市</th>
      <th>會員描述</th>
      <th class="col-sm-1">詳細</th>
    </tr>
  </thead>
  <tbody>
    @foreach (var mb in Model)
    {
      <tr class="row">
        <td class="col-sm-2">@mb.MemberName</td>
        <td>@mb.Email</td>
        <td>@mb.CityName</td>
        <td>@mb.Description</td>
        <td class="col-sm-1">@Html.ActionLink("詳細", "MemberInfo", new { id = mb.MemberID })</td>
      </tr>
    }
  </tbody>
</table>
```

ViewModel -6

- Controller：將查詢到的資料填入至Model。

```
public ActionResult MemberList()
{
    IEnumerable<MembersViewModel> qMemberList = db.Members.Select(m =>
        new MembersViewModel()
        {
            MemberID = m.MemberID,
            MemberName = m.MemberName,
            Email = m.Email,
            Description = m.Description,
            CityName = m.City.CityName
        });
    return View(qMemberList);
}
```

ViewModel -7

MemberList

新增會員資料

會員姓名	Email	城市	會員描述	編輯
John Wick	johnwick@jw.com	New York	John is a man of focus, commitment, sheer will... something you know very little about. I once saw him kill three men in a bar... with a pencil. With a fucking pencil!"	詳細
張滿月	JangMan-wol@delluna.com	Seoul	호텔 델루나 이지은	詳細
王雷	WangLei@ganlingnaa.com	Singapore	早上買，中午吃，晚上走	詳細
諸葛村夫	farmer@shu.com	Sichuan	臣本布衣，躬耕於南陽，苟全性命於亂世，不求聞達於諸侯。先帝不以臣卑鄙，猥自枉屈，三顧臣於草廬之中，諮臣以當世之事，由是感激，遂許先帝以驅馳。後值傾覆，受任於敗軍之際，奉命於危難之間：爾來二十有一年矣。	詳細
陳浩南	AndyChan@teddyboy.com	Hong Kong	銅鑼灣只有一個浩南，就是我陳浩南!	詳細

EF Core指令

- 先裝套件（專案要用到）：

```
dotnet add package Microsoft.EntityFrameworkCore.SqlServer  
dotnet add package Microsoft.EntityFrameworkCore.Tools
```

```
dotnet tool install --global dotnet-ef
```

- Scaffold 反向產生（Database First 核心指令）：**

```
dotnet ef dbcontext scaffold  
"Server=.;Database=YourDb;Trusted_Connection=True;Encrypt=True;TrustServerCertificate=True"  
Microsoft.EntityFrameworkCore.SqlServer -o Models -c YourDbContext -f
```

- **-o Models**：實體類別放到 Models/
- **-c YourDbContext**：DbContext 命名
- **-f**：覆蓋既有檔案（重新 scaffold 時必備）